

UE Apprentissage SIA S9

Supervise Learning

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Outline

Introduction

Introductory Example

Multi layer Perceptron

Convolutional Neural Networks

Transformer

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Introductory Example

- Linear model for regression

- Linear model for classification

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Convolutional Neural Networks

- Concepts

- How to improve

Transformer

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- CV:

2004-2007 Ph.D. degree in Signal and Image Processing from the INP, Grenoble & the University of Iceland

2007-2008 Assistant Professor Grenoble

2008-2010 Post-doc position at INRIA - MISTIS Team

2010-2011 Assistant Professor Toulouse

2011-2018 Associate Professor at DYNAFOR & INP, Toulouse

Since 2018 Research (CRCN) at CESBIO, INRAe

- Research interests are:

Machine learning for environmental/ecological monitoring

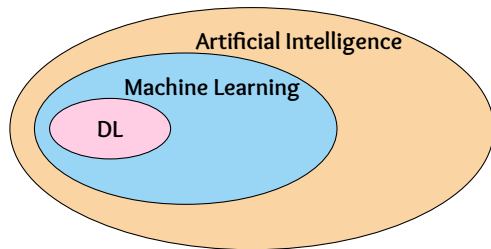
Course Objectives

TBC

What is Supervised (Machine) Learning ?

- **Artificial Intelligence**

Perform human tasks using computers and algorithms



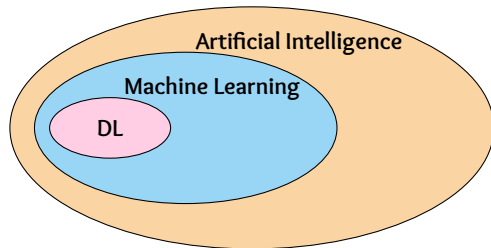
What is Supervised (Machine) Learning ?

- **Artificial Intelligence**

Perform human tasks using computers and algorithms

- Mitchell, *The discipline of machine learning:*

Machine Learning is defined as the capacity of a computer program to improve its performance measure with observations



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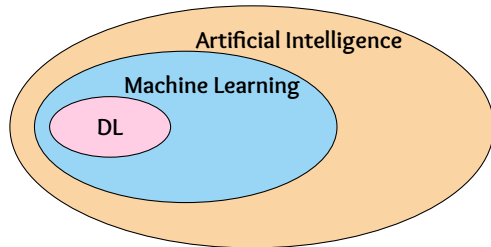
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- **Deep Learning**

- ★ Lot of data data
- ★ Complex model
- ★ High performance computing



Main Notations

Data

- Observed data $\mathbf{x} \in \mathbb{R}^d$, called *input variables* or *predictors*.
- Data to be predicted $y \in \mathbb{R}^p$, called *output variables* or *responses*.

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Learning Problem

- If y are quantitative data: *regression* and $y \in \mathbb{R}^p$.
- If y are categorical data: *classification* and $y \in \{C_1, \dots, C_C\}$ with C_i refers to the i^{th} category.

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Prediction function

$$\begin{aligned} f_{\theta} : \mathbb{R}^d &\rightarrow \mathbb{R}^p \\ \mathbf{x} &\mapsto y \end{aligned}$$

Online References

- Aston Zhang et al. “Dive into Deep Learning”. In: *arXiv preprint arXiv:2106.11342* (2021)
- Simon J.D. Prince. *Understanding Deep Learning*. MIT Press, 2023. URL: <https://udlbook.github.io/udlbook/>
- Kevin P. Murphy. *Probabilistic Machine Learning: An introduction*. MIT Press, 2022. URL: <http://probml.github.io/book1>
- Kevin P. Murphy. *Probabilistic Machine Learning: Advanced Topics*. MIT Press, 2023. URL: <http://probml.github.io/book2>

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Univariate linear model

- $f(x) = wx + b$ and $\theta = (w, b)$
- Loss function: $\ell(f(x_i), y_i) = (f(x_i) - y_i)^2$
- Objective function:

$$\mathcal{L} = \frac{1}{n} \sum_{i=1}^n \ell_i$$

- Gradients update: $\theta^{t+1} = \theta^t - \eta \nabla_{\theta} \mathcal{L}$

$$\frac{\partial \mathcal{L}}{\partial w} = \frac{2}{n} \sum_{i=1}^n x_i (wx_i + b - y_i)$$

$$\frac{\partial \mathcal{L}}{\partial b} = \frac{2}{n} \sum_{i=1}^n (wx_i + b - y_i)$$

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Work

Implement the 1D regression in pytorch of the following function:

$$f(x) = 2x - 1$$

Do the notebook:

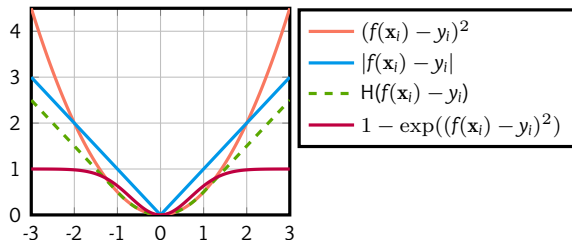
[regression_1d_toy.ipynb](#)

Multivariate linear model

- $f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b$ and $\boldsymbol{\theta} = (\mathbf{w}, b)$
- Same loss and objective function and so same gradient updates ...

Multivariate linear model

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- We can use other loss function

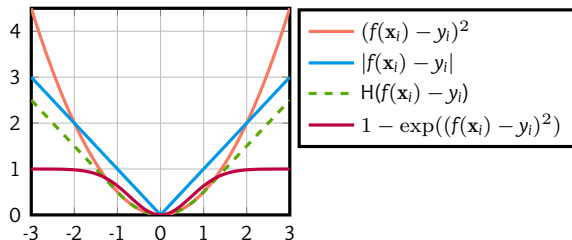


with H stands for *Hubert loss*:

$$H(f(\mathbf{x}_i) - y_i) = \begin{cases} \frac{1}{2}(f(\mathbf{x}_i) - y_i)^2 & \text{for } |f(\mathbf{x}_i) - y_i| \leq \delta, \\ (|f(\mathbf{x}_i) - y_i| - 0.5), & \text{otherwise.} \end{cases}$$

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Work

On the **California housing** data set

- Prepare (train/validation) data
- Define a multi. linear model
- Optimize your hyperparameters (batch size, learning rate)
- Switch L2 to other loss function

Scaling feature for multivariate linear model with L2 loss function

- Suppose we have two features of different scale (e.g. because of different unit)

$$\mathbf{x}_1 \sim \mathcal{N}(0, 1) \text{ and } \mathbf{x}_2 \sim \mathcal{N}(10, 10)$$

- What is the impact on the gradient update ?

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- Noting $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n]^\top$ and $\mathbf{y} = [y_1, \dots, y_n]^\top$

$$\nabla_{\mathbf{w}} = -\mathbf{X}^\top \underbrace{(\mathbf{y} - \mathbf{X}\mathbf{w})}_{\mathbf{e}} \Rightarrow \nabla_{\mathbf{w}_p} = -\sum_{i=1}^n \mathbf{e}_i \mathbf{x}_{ip}$$

- Parameter update

$$\mathbf{w}_p^{(t+1)} = \mathbf{w}_p^{(t)} + \frac{\eta}{n} \sum_{i=1}^n \mathbf{e}_i^{(t)} \mathbf{x}_{ip}$$

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- Standardization of the input feature (and for regression the output too): [scaling_feature.ipynb](#)

$$\tilde{\mathbf{x}} = (\mathbf{x} - \boldsymbol{\mu}) \oslash \boldsymbol{\sigma}$$

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Classification as a regression problem

- Predict categorical variables ("Dogs", "Cats", "Monkey" ...)
- Estimation of the posterior probability

$$p(y_i = c | \mathbf{x}_i) \forall c \in \{1, \dots, C\} \text{ with } p(y_i = c | \mathbf{x}_i) \geq 0 \text{ and } \sum_{c=1}^C p(y_i = c | \mathbf{x}_i) = 1$$

- MAP rule: $\hat{c}_i = \arg \max_c p(y_i = c | \mathbf{x}_i)$

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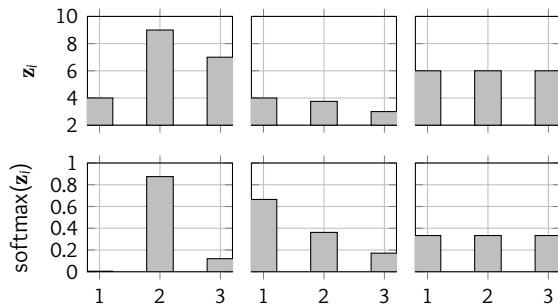
- MAP rule: $\hat{c}_i = \arg \max_c p(y_i = c | \mathbf{x}_i)$
- Softmax linear regression

$$1 = \sum_{c=1}^C p(y_i = c | \mathbf{x}_i) = \sum_{c=1}^C \exp \left\{ \mathbf{w}_c^\top \mathbf{x}_i + b_c + K_i \right\} = \exp \{K_i\} \sum_{c=1}^C \exp \left\{ \mathbf{w}_c^\top \mathbf{x}_i + b_c \right\}$$

thus

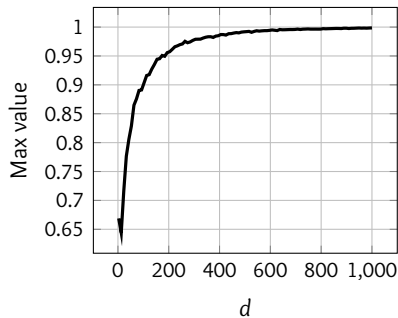
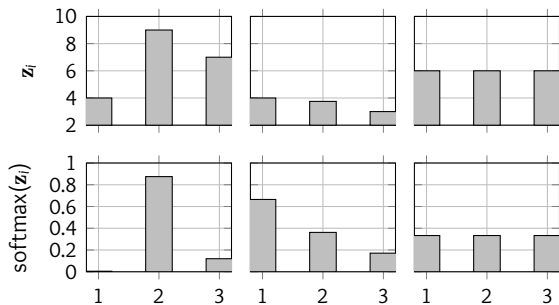
$$p(y_i = c | \mathbf{x}_i) = \frac{\exp \left\{ \mathbf{w}_c^\top \mathbf{x}_i + b_c \right\}}{\sum_{c'=1}^C \exp \left\{ \mathbf{w}_{c'}^\top \mathbf{x}_i + b_{c'} \right\}} = \frac{\exp(\mathbf{z}_{ic})}{\sum_{c'=1}^C \exp(\mathbf{z}_{ic'})} = \text{softmax}(\mathbf{z}_i)_c$$

Softmax



- Translation invariant: $\text{softmax}(\mathbf{z} + K) = \text{softmax}(\mathbf{z})$
- Preserve ordering relation: $\arg \max_c(\mathbf{p}_i) = \arg \max_c(\mathbf{z}_i)$
- If $\mathbf{z} \in \mathbb{R}^d$ with $d \rightarrow \infty$ then $\text{softmax}(\mathbf{z}) \rightarrow \text{"One-hot-vector"}$

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- If $\mathbf{z} \in \mathbb{R}^d$ with $d \rightarrow \infty$ then $\text{softmax}(\mathbf{z}) \rightarrow \text{"One-hot-vector"}$

Cross entropy loss function

- One hot encoding: $y_i = c \rightarrow \mathbf{y}_i \in \mathbb{R}^C$, $\mathbf{y}_{ic} = 1$ if $i = c$ else 0
- Likelihood:

$$l(\mathbf{y}_i, \hat{\mathbf{y}}_i) = \prod_{c'=1}^C p(y_i = c' | \mathbf{x}_i)^{\mathbf{y}_{ic'}} = \frac{\exp(\mathbf{z}_{ic})}{\sum_{c'=1}^C \exp(\mathbf{z}_{ic'})}$$

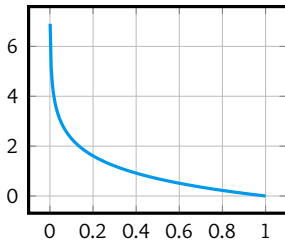
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- Negative log likelihood:

$$\begin{aligned}\ell(\mathbf{y}_i, \hat{\mathbf{y}}_i) &= - \sum_{c'=1}^C y_{ic'} \log \{p(y_i = c' | \mathbf{x}_i)\} \\ &= -\log \{p(y_i = c | \mathbf{x}_i)\} \\ &= -(\mathbf{w}_c^\top \mathbf{x}_i + b_c) + \underbrace{\log \left\{ \sum_{c'=1}^C \exp(\mathbf{w}_{c'}^\top \mathbf{x}_i + b_{c'}) \right\}}_{\text{logsumexp}(\mathbf{z}_i)}\end{aligned}$$



Work

What happens if:

- All $\mathbf{z}_{ic}$ are small ?
- One $\mathbf{z}_{ic}$ is very large ?
- All $\mathbf{z}_{ic}$ are very large ?

Multivariate linear classifier

- The model

$$\mathbf{p}_i = \text{softmax}(\mathbf{W}\mathbf{x}_i + \mathbf{b})$$

- Cross entropy loss function
- Same comments than for regression
 - ★ Split train, val and test
 - ★ Standardization
 - ★ Batch training
 - ★ Tune learning rate

Work

Implement a linear classifier in pytorch for the classification of **Fashion MNIST** data set.

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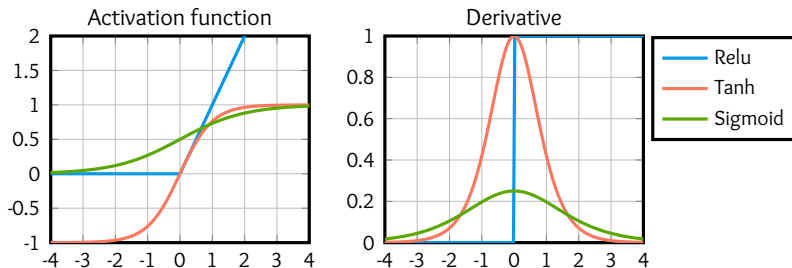
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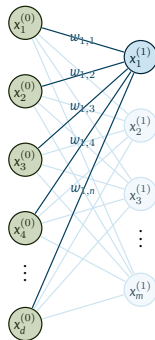
Activation function

- As discussed earlier we need non-linearity between two layers
- Done with *activation function* between two layers
 - ★ Rectified Linear (ReLU): $\max(x, 0)$
 - ★ Sigmoid: $\frac{1}{1 + \exp(-x)}$
 - ★ Tanh: $\tanh(x)$



Multi Layer Perceptron (MLP)

- $\mathbf{x}^{(1)} = \sigma(\mathbf{W}^{(0)}\mathbf{x}^{(0)} + \mathbf{b}^{(0)})$
- $\mathbf{x}^{(2)} = \sigma(\mathbf{W}^{(1)}\mathbf{x}^{(1)} + \mathbf{b}^{(1)})$
- ...
- $\mathbf{y} = \mathbf{W}^H\mathbf{x}^H + \mathbf{b}^H$



$$= \sigma(w_{1,0}x_0^{(0)} + w_{1,1}x_1^{(0)} + \dots + w_{1,d}x_d^{(0)} + b_1^{(0)})$$

$$= \sigma\left(\sum_{i=1}^d w_{1,i}x_i^{(0)} + b_1^{(0)}\right)$$

$$\begin{pmatrix} x_1^{(1)} \\ x_2^{(1)} \\ \vdots \\ x_m^{(1)} \end{pmatrix} = \sigma \left[\begin{pmatrix} w_{1,0} & w_{1,1} & \dots & w_{1,d} \\ w_{2,0} & w_{2,1} & \dots & w_{2,d} \\ \vdots & \vdots & \ddots & \vdots \\ w_{m,0} & w_{m,1} & \dots & w_{m,d} \end{pmatrix} \begin{pmatrix} x_1^{(0)} \\ x_2^{(0)} \\ \vdots \\ x_d^{(0)} \end{pmatrix} + \begin{pmatrix} b_1^{(0)} \\ b_2^{(0)} \\ \vdots \\ b_m^{(0)} \end{pmatrix} \right]$$

$$\mathbf{x}^{(1)} = \sigma(\mathbf{W}^{(0)}\mathbf{x}^{(0)} + \mathbf{b}^{(0)})$$

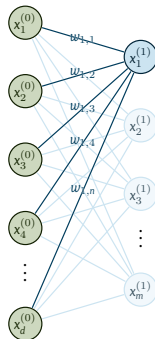
From https://tikz.net/neural_networks/

Multi Layer Perceptron (MLP)

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- $\mathbf{x}^{(2)} = \sigma(\mathbf{W}^{(1)}\mathbf{x}^{(1)} + \mathbf{b}^{(1)})$
- ...
- $\mathbf{y} = \mathbf{W}^H\mathbf{x}^H + \mathbf{b}^H$

Work

- For an univariate regression problem, shows that a 3-layers MLP is a piece-wise linear function.
- Implement an MLP classifier for Fashion MNIST data set



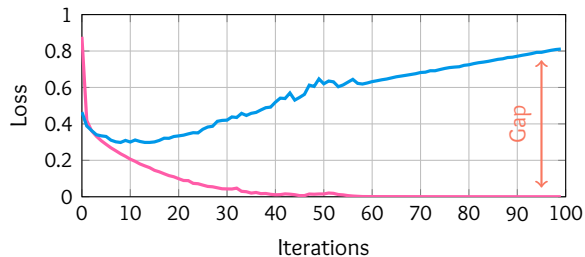
$$\begin{aligned} &= \sigma(w_{1,0}x_0^{(0)} + w_{1,1}x_1^{(0)} + \dots + w_{1,d}x_d^{(0)} + b_1^{(0)}) \\ &= \sigma\left(\sum_{i=1}^d w_{1,i}x_i^{(0)} + b_1^{(0)}\right) \end{aligned}$$

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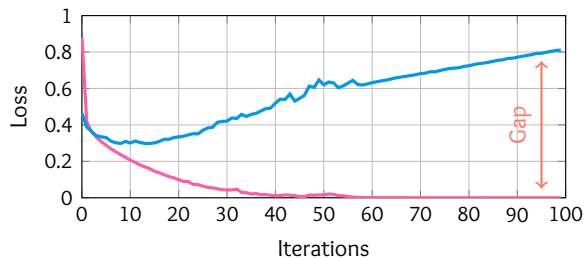
Generalization Error

- *Generalization error*: Difference between the train accuracy and val/test accuracy
- *Overfitting*: High generalization error



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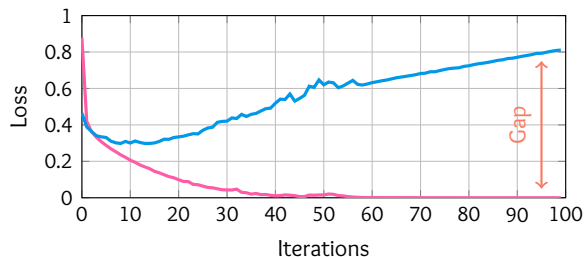
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Theory: favor simpler model and/or smooth model

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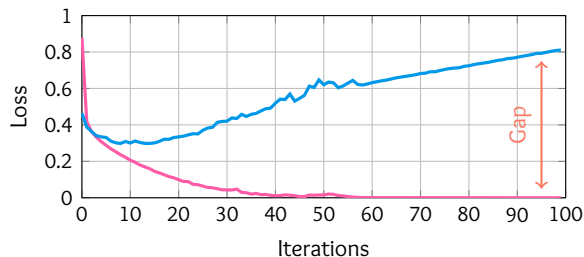


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- Early stopping: monitor the validation loss

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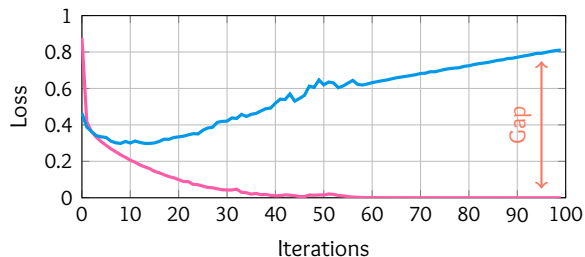
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$$\mathcal{L} = \frac{1}{n} \sum_{i=1}^n \ell_i + \frac{\lambda}{H} \sum_{h=1}^H \|\mathbf{w}_h\|_2^2$$

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Theory: favor simpler model and/or smooth model

- Early stopping: monitor the validation loss
- Weight decay: Tikhonov regularization
- Noise injection: Dropout

$$\mathcal{L} = \frac{1}{n} \sum_{i=1}^n \ell_i + \frac{\lambda}{H} \sum_{h=1}^H \|\mathbf{w}_h\|_2^2$$

Dropout

- Smoothness: $f(\mathbf{x} + \epsilon) \approx f(\mathbf{x})$
- Idea: Noise injection during the learning process such as $\mathbb{E}[\tilde{\mathbf{x}}] = \mathbf{x}$
- With MLP:

$$f(\mathbf{x}) = f_H \circ f_{H-1} \circ \dots \circ f_0(\mathbf{x})$$

$$\text{and } \mathbf{x}^{(h+1)} = f_h(\mathbf{x}^{(h)}) = \sigma(\mathbf{W}^{(h)}\mathbf{x}^{(h)} + \mathbf{b}^{(h)})$$

- Dropout: remove (set to zero) some internal features with probability p

$$\mathbf{x}^{(h+1)} = f_h(\tilde{\mathbf{x}}^{(h)}) = \sigma(\mathbf{W}^{(h)}(\mathbf{m}^{(h)} \odot \mathbf{x}^{(h)}) + \mathbf{b}^{(h)})$$

with

$$\mathbf{m}_i^{(h)} = \begin{cases} 0 & \text{with probability } p \\ 1/(1-p) & \text{with probability } 1-p \end{cases}$$

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- Enable during training and disable during inference (can be used to estimate posterior distribution)

$$f(\tilde{\mathbf{x}}) = f_H \circ D_H \circ f_{H-1} \circ \dots \circ D_1 \circ f_0(\mathbf{x})$$

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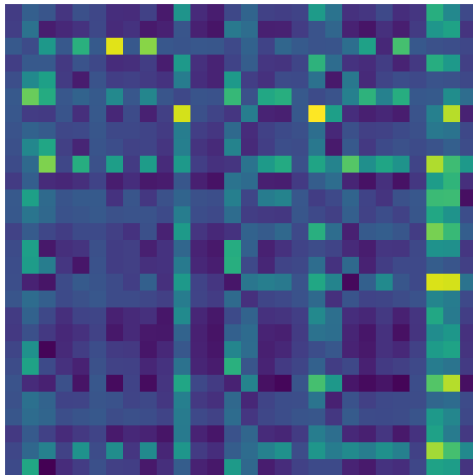
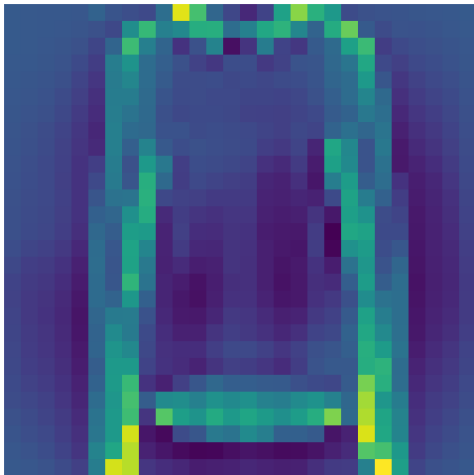
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Image are not a vector of pixels



2D Convolution

- Fully connected layer:

$$x_{pq}^{(1)} = \sigma \left(\sum_{i,j=1}^{H,W} \mathbf{w}_{pqij}^{(0)} x_{ij}^{(0)} + b_{pq}^{(0)} \right)$$

- Remove dependence w.r.t to pixel position $\approx$ translation invariant

$$x_{pq}^{(1)} = \sigma \left(\sum_{i,j=1}^{H,W} \mathbf{w}_{ij}^{(0)} x_{ij}^{(0)} + b^{(0)} \right)$$

- Act locally: restrict to neighborhood of pixel p, q

$$x_{pq}^{(1)} = \sigma \left(\sum_{i,j=-\Delta}^{i,j=+\Delta} \mathbf{w}_{ij}^{(0)} x_{(p+i)(q+j)}^{(0)} + b^{(0)} \right)$$

$$\begin{pmatrix} w_{pq11} & w_{pq12} & \dots & w_{pq1j} & \dots & w_{pq1W} \\ w_{pq21} & w_{pq22} & \dots & w_{pq2j} & \dots & w_{pq2W} \\ \vdots & \vdots & & \vdots & & \vdots \\ w_{pqi1} & w_{pqi2} & \dots & w_{pqij} & \dots & w_{pqiW} \\ \vdots & \vdots & & \vdots & & \vdots \\ w_{pqH1} & w_{pqH2} & \dots & w_{pqHj} & \dots & w_{pqHW} \end{pmatrix}$$

2D Convolution

- Fully connected layer:

$$x_{pq}^{(1)} = \sigma \left(\sum_{i,j=1}^{H,W} \mathbf{w}_{pqij}^{(0)} x_{ij}^{(0)} + b_{pq}^{(0)} \right)$$

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$$\begin{pmatrix} 0 & \dots & 0 & \dots & 0 \\ 0 & \dots & 0 & \dots & 0 \\ \vdots & w_{(i)(j-1)} & w_{(i-1)(j)} & w_{(i-1)(j+1)} & \vdots \\ 0 & w_{(i)(j-1)} & w_{(i)(j)} & w_{(i)(j+1)} & 0 \\ \vdots & w_{(i+1)(j-1)} & w_{(i+1)(j)} & w_{(i+1)(j+1)} & \vdots \\ 0 & \dots & 0 & \dots & 0 \end{pmatrix}$$

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Work

Show that 2D convolution can be expressed as matrix multiplication.

Basics on 2D Convolution 1/2

Convolution

0	1	2	3	4
5	6	7	8	9
10	11	12	13	14
15	16	17	18	19

$\chi^{(h)}$

*

1	1	1
1	1	1
1	1	1

K

=

54	63	72
99	108	117

$\chi^{(h+1)}$

Basics on 2D Convolution 1/2

Convolution + Padding (1,1)

0	0	0	0	0	0	0
0	0	1	2	3	4	0
0	5	6	7	8	9	0
0	10	11	12	13	14	0
0	15	16	17	18	19	0
0	0	0	0	0	0	0

*

1	1	1
1	1	1
1	1	1

=

12	21	27	33	24
33	54	63	72	51
63	99	108	117	81
52	81	87	93	64

Basics on 2D Convolution 1/2

Convolution + Padding (1,1) + Stride (3, 2)

0	0	0	0	0	0	0
0	0	1	2	3	4	0
0	5	6	7	8	9	0
0	10	11	12	13	14	0
0	15	16	17	18	19	0
0	0	0	0	0	0	0

1	1	1
1	1	1
1	1	1

12	27	24
52	87	64

Basics on 2D Convolution 1/2

Convolution for multivalued images

0	1	2	3	4
5	6	7	8	9
10	11	12	13	14
15	16	17	18	19

*

1	1	1
1	1	1
1	1	1

=

54	63	72
99	108	117

+

=

288	279	270
243	234	225

19	18	17	16	15
14	13	12	11	10
9	8	7	6	5
4	3	2	1	0

*

2	2	2
2	2	2
2	2	2

=

234	216	198
144	126	108

Basics of 2D Convolution 2/2

- Multivariate input - Multivariate output

$$\begin{aligned}\mathbf{X}^{(h)} &\in \mathbb{R}^{C_i \times H^{(h)} \times W^{(h)}} \\ \mathbf{K}^{(h)} &\in \mathbb{R}^{C_o \times C_i \times k_H^h \times k_W^h} \\ \mathbf{X}^{(h+1)} &\in \mathbb{R}^{C_o \times H^{(h+1)} \times W^{(h+1)}}\end{aligned}$$

- Special case of 1×1 convolution: $(C_o, C_i, k_H, k_W) = (C_o, C_i, 1, 1)$
 - ★ Combine pixel across channels - no spatial operation
 - ★ Matrix product / linear layer in the channel dimension

$$\mathbf{X}^{(h+1)} = \text{reshape} \left\{ \sigma(\mathbf{K}^{(h)} \text{reshape} \{\mathbf{X}^{(h)}, C_i, HW\} + \mathbf{b}^{(h)}), C_o, H, W \right\}$$

with $\text{reshape} \{\mathbf{X}^{(h)}, C_i, HW\} \in \mathbb{R}^{C_i \times HW}$

Pooling

- Reduce the resolution of the image $\equiv$ Use larger convolution kernel
- Downsampling with simple operator on the neighborhood:
 - ★ Average value
 - ★ Max value (preferred)
- After the non-linearity

0	1	2	3	4
5	6	7	8	9
10	11	12	13	14
15	16	17	18	19

2×2 MaxPooling
Stride=2

6	8
16	18

Work

- Express multi-channels/outputs as matrix product
- Show that the successive convolution with two kernels is equivalent to one convolution
- Show that $\max(x,y)$ can be written using 1D-Convolution and ReLU
- What kind of padding will you use with max pooling ?
- Implement a CNN for the classification of *Fashion MNIST* data set: use one or two convolutional layer for the MLP.

Outline

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Introductory Example

- Linear model for regression

- Linear model for classification

Multi layer Perceptron

Convolutional Neural Networks

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